

Monocular Depth Estimation: Model Survey for RPX Benchmark

Survey date: 2025-05-07

Scope: State-of-the-art monocular metric and relative depth estimation models. The RPX benchmark covers 99 scenes (64 indoor + 35 outdoor/transitional) captured with a D435+T265+GoPro rig, spanning tabletop manipulation, atriums, staircases, gardens, tennis courts, and fountains.

Criterion: Open-source code **and** publicly downloadable model checkpoints required for inclusion in the benchmark.

1. Models Included in Benchmark

The following models have verified open-source implementations and publicly available pretrained weights as of the survey date.

1.1 UniDepth V2

Field	Details
Paper	“UniDepthV2: Universal Monocular Metric Depth Estimation Made Simpler”
Venue	arXiv:2502.20110 (Feb 2025); UniDepth V1 at CVPR 2024 Highlight
Affiliation	ETH Zurich
Code	https://github.com/lpiccinelli-eth/UniDepth
Checkpoints	<code>lpiccinelli/unidepth-v2-vitl14</code> (ViT-L, 3.4M downloads/month), <code>lpiccinelli/unidepth-v2-vits14</code> (ViT-S) on HuggingFace
License	See repo
Output	Metric depth (meters). Also predicts camera intrinsics – no intrinsics input required.
Key claim	Overall new SOTA in zero-shot metric monocular depth estimation; ranks #1 among published methods on KITTI Depth Prediction benchmark.
Relevance	Camera-agnostic metric prediction is directly useful for robotics where camera parameters may vary or be imprecise.

1.2 Depth Pro

Field	Details
Paper	“Depth Pro: Sharp Monocular Metric Depth in Less Than a Second”
Venue	ICLR 2025
Affiliation	Apple
Code	https://github.com/apple/ml-depth-pro
Checkpoints	<code>apple/DepthPro</code> (original), <code>apple/DepthPro-hf</code> (transformers-native) on HuggingFace
License	Apple Sample Code License
Output	Metric depth at 2.25 megapixel resolution. No camera intrinsics needed.
Key claim	0.3 seconds per frame on a standard GPU. Sharp high-frequency boundary details.
Relevance	Speed + boundary sharpness are critical for real-time robotics perception of object edges.

1.3 Depth Anything V2

Field	Details
Paper	“Depth Anything V2”
Venue	NeurIPS 2024
Affiliation	HKU, TikTok
Code	https://github.com/DepthAnything/Depth-Anything-V2
Checkpoints (relative)	<code>depth-anything/Depth-Anything-V2-Small-hf</code> (24.8M params, 1.8M dl/mo), <code>-Base-hf</code> (97.5M), <code>-Large-hf</code> (335.3M)
Checkpoints (metric)	<code>depth-anything/Depth-Anything-V2-Metric-Hypersim-Large</code> (indoor-trained), <code>Depth-Anything-V2-Metric-VKITTI-Large</code> (outdoor-trained)
License	Apache 2.0
Output	Relative depth (base models); metric depth (fine-tuned variants).
Key claim	Trained on 595K synthetic + 62M real unlabeled images. Finer detail and more robust than V1. 10x faster than diffusion-based methods.
Relevance	Both metric variants benchmarked: Hypersim-trained (indoor-focused) and VKITTI-trained (outdoor-focused). RPX has both indoor and outdoor scenes, so this directly tests domain-gap effects. Multiple model sizes enable latency/accuracy tradeoff analysis.

1.4 Metric3D v2

Field	Details
Paper	“Metric3D v2: A Versatile Monocular Geometric Foundation Model for Zero-shot Metric Depth and Surface Normal Estimation”
Venue	IEEE TPAMI 2024
Affiliation	Multiple (see paper)
Code	https://github.com/YvanYin/Metric3D
Checkpoints	Auto-downloaded via repo scripts
License	See repo
Output	Metric depth + surface normals jointly.
Key claim	Zero-shot metric depth with auxiliary normal estimation improves geometric accuracy.
Relevance	Joint depth + normal output is valuable for robotics grasp planning and surface reasoning.

1.5 MoGe-2

Field	Details
Paper	“MoGe-2: Accurate Monocular Geometry with Metric Scale and Sharp Details”
Venue	NeurIPS 2025 (MoGe v1: CVPR 2025 Oral)
Affiliation	Microsoft Research
Code	https://github.com/microsoft/MoGe
Checkpoints	<code>Ruicheng/moge-2-vitl-normal</code> on HuggingFace. Training code released Oct 2025.
License	See repo
Output	Metric-scale 3D point map + normal map from a single image.

Field	Details
Key claim	Extends MoGe (affine-invariant point maps) to absolute metric scale with improved sharpness and lower latency.
Relevance	Full 3D point map output (not just depth) is directly useful for robotic manipulation planning.

1.6 MetricSolver

Field	Details
Paper	“Metric-Solver: Sliding Anchored Metric Depth Estimation from a Single Image”
Venue	arXiv:2504.12103 (Apr 2025)
Affiliation	Tele-AI
Code	https://github.com/Tele-AI/MetricSolver
Checkpoints	Provided in repo
License	MIT
Output	Metric depth with dynamic scale anchoring.
Key claim	Sliding anchor method adapts to diverse depth scales across indoor/outdoor without separate heads.
Relevance	Handles the scale diversity present in robotics scenes (close tabletop objects vs. room-scale).
Note	ICLR 2026 submission was withdrawn; code and weights remain publicly available.

1.7 Marigold

Field	Details
Paper	“Repurposing Diffusion-Based Image Generators for Monocular Depth Estimation”
Venue	CVPR 2024 (Oral, Best Paper Award Candidate)
Affiliation	ETH Zurich
Code	https://github.com/prs-eth/Marigold
Checkpoints	Multiple variants on HuggingFace under prs-eth/
License	Apache 2.0
Output	Relative depth (affine-invariant). Requires post-hoc alignment for metric evaluation.
Key claim	Leverages Stable Diffusion visual priors. Strong zero-shot generalization from synthetic-only fine-tuning.
Relevance	Diffusion-based baseline representing a different architectural paradigm.

1.8 Lotus / Lotus-2

Field	Details
Paper	“Lotus: Diffusion-based Visual Foundation Model for High-quality Dense Prediction”
Venue	ICLR 2025 (Lotus-2: arXiv:2512.01030, Nov 2025)
Affiliation	HKUST
Code	https://github.com/chenxwh/Lotus

Field	Details
Checkpoints	jingheya/lotus-depth-g-v2-1-disparity (Lotus v2, 4.2K dl/mo), jingheya/Lotus-2 (Lotus-2 with sharpener + normals)
License	See repo
Output	Relative depth (disparity space). Also normal estimation in Lotus-2.
Key claim	Single-step prediction (not iterative diffusion), faster than Marigold with comparable or better quality.
Relevance	Faster diffusion-family alternative; Lotus-2 adds boundary sharpening relevant to object segmentation.

1.9 ZoeDepth

Field	Details
Paper	“ZoeDepth: Zero-shot Transfer by Combining Relative and Metric Depth”
Venue	arXiv:2302.12288 (2023)
Affiliation	Intel
Code	https://github.com/isl-org/ZoeDepth
Checkpoints	Provided in repo
License	MIT
Output	Metric depth.
Key claim	Combines relative depth backbone with metric depth heads for domain-specific transfer.
Relevance	Legacy baseline. Repository archived May 2025 – still functional but no longer maintained.

2. Models NOT Benchmarked – Unavailable Code/Checkpoints

The following recent models are relevant but **cannot be included** because open-source code and/or pretrained checkpoints were not publicly available as of **2025-05-07**.

Model	Paper	Venue	Reason for Exclusion
AsyncMDE	“Real-Time Monocular Depth Estimation via Asynchronous Spatial Memory”	arXiv:2603.10438 (Mar 2026)	No code or model weights released.
RTS-Mono	“A Real-Time Self-Supervised Monocular Depth Estimation Method for Real-World Deployment”	arXiv:2511.14107 (Nov 2025)	No code or model weights released.
PTC-Depth	“Pose-Refined Monocular Depth Estimation with Temporal Consistency”	arXiv:2604.01791 (Apr 2026)	No code or model weights released.

Model	Paper	Venue	Reason for Exclusion
AnyDepth	“Depth Estimation Made Easy” (DINOv3 encoder)	arXiv:2601.02760 (Jan 2026)	No code or model weights released.
KineDepth	“Utilizing Robot Kinematics for Online Metric Depth Estimation”	IROS 2025 (arXiv:2409.19490)	Implementation tightly coupled to a specific robot platform; not generalizable without hardware.

Statement for Reviewers

These models represent relevant concurrent or subsequent work. Their exclusion from our benchmark results is solely due to the absence of publicly available code and pretrained weights at the time of evaluation (2025-05-07). Our benchmark dataset (75K frames across 99 indoor and outdoor scenes) and evaluation toolkit are publicly released; **we invite the authors of these methods to evaluate on our dataset**, which demonstrates the self-sustaining nature of our benchmark – new methods can be evaluated against the same standardized data and metrics without our direct involvement.

3. Additional Context

3.1 Evaluation Protocol Reference

- **4th Monocular Depth Estimation Challenge (MDEC)**, CVPR 2025 Workshop (arXiv:2504.17787): switched to least-squares alignment (from median). 24 submissions.
- **BenchDepth** (<https://zhyever.github.io/benchdepth>): proposes alignment-free evaluation of depth foundation models through five carefully designed tasks.
- **“How Should One Evaluate Monocular Depth Estimation?”** (arXiv:2510.19814): critiques standard metrics (delta1, AbsRel) as insensitive to curvature perturbations.

3.2 Video/Temporal Depth (out of scope for this benchmark phase)

- **DepthCrafter** (CVPR 2025 Highlight, Tencent): temporally consistent video depth. Code at <https://github.com/Tencent/DepthCrafter>. Relevant if the benchmark expands to sequential frames.

3.3 Deferred Models (noted in CHANGELOG.md)

Per the RPX benchmark CHANGELOG, these are explicitly deferred: - **Video Depth Anything** – needs temporal eval mode - **Prompt Depth Anything** – needs sparse-depth prompt task - **Depth Anything 3** – not yet in `transformers` library

4. Summary Table

4.1 Metric Depth Models (10)

#	Model	Venue	Code	Checkpoints	Notes
1	UniDepth V2	CVPR’24 / arXiv’25	github	lpiccinelli/unidepth-v2-vit19	19 TA metric, no intrinsics
2	Depth Pro	ICLR 2025	github	apple/DepthPro-hf	Sharp edges, 0.3s, Apple

#	Model	Venue	Code	Checkpoints	Notes
3	Depth Anything V2	NeurIPS 2024	github	depth-anything/Depth-Anything-V2	Large Metric-Hypersim-Large (Hypersim-trained)
4	Metric-Indoor Depth Anything V2 Metric-Outdoor	NeurIPS 2024	same repo	depth-anything/Depth-Anything-V2-Metric-VKITTI-Large	(VKITTI-trained) – domain-gap probe across RPX’s indoor+outdoor mix
5	Metric3D v2	TPAMI 2024	github	In repo (auto-download)	Joint depth + normals
6	MoGe-2	NeurIPS 2025	github	Ruicheng/moge-2-vitl-normal	Metric 3D point map + normals
7	MetricSolver	arXiv Apr 2025	github	In repo	Sliding anchor, MIT license
8	ZoeDepth	arXiv 2023	github	In repo	Legacy baseline, archived May 2025
9	PatchFusion	CVPR 2024	github	zhyever/patchfusion_zoedepth + variants	Tile-based high-res metric, MIT
10	HyDen (MetaDepth)	ICLR 2026	github	facebook/hyden-da2-metric-deptha	Deptha, hybrid dual-path encoder

4.2 Relative Depth Models (10)

#	Model	Venue	Code	Checkpoints	Notes
1	Depth Anything V2 (relative)	NeurIPS 2024	github	depth-anything/Depth-Anything-V2-Relative-hf (335M)	Large
2	Depth Anything V1	CVPR 2024	github	LiheYoung/depth-anything-large-hf	Baseline
3	Marigold v1.1	CVPR 2024 Oral	github	prs-eth/marigold-depth-v1-1	Diffusion-based, best paper candidate
4	Marigold LCM	CVPR 2024 Oral	same repo	prs-eth/marigold-depth-lcm-v1-0	Fluxer variant (LCM distillation)
5	Lotus-2	ICLR 2025	github	jingheya/Lotus-2	Single-step diffusion + normals
6	GeoWizard	ECCV 2024	github	On HuggingFace	Joint depth + normals, diffusion
7	MiDaS v3.1 (DPT-BEiT-L)	TPAMI 2022	github	Intel/dpt-beit-large-384	Classic baseline, 5 backbone variants
8	Distill-Any-Depth	arXiv Feb 2025	github	xingyang1/Distill-Any-Depth-Variants-hf (61K dl/mo)	Multi-teacher distillation of DA-V2

#	Model	Venue	Code	Checkpoints	Notes
9	MoGe (v1)	CVPR 2025 Oral	github	Ruicheng/moge-vit1	Affine-invariant 3D point map
10	HyDen (relative)	ICLR 2026	github	facebook/hyden-da2-relative-depth	relative-depth variant

4.3 Models NOT Benchmarked – Unavailable Code/Checkpoints

Model	Paper	Venue	Reason for Exclusion
AsyncMDE	arXiv:2603.10438	arXiv Mar 2026	No code or weights released
RTS-Mono	arXiv:2511.14107	arXiv Nov 2025	No code or weights released
PTC-Depth	arXiv:2604.01791	arXiv Apr 2026	No code or weights released
AnyDepth	arXiv:2601.02760	arXiv Jan 2026	No code or weights released
KineDepth	arXiv:2409.19490	IROS 2025	Code tightly coupled to specific robot platform